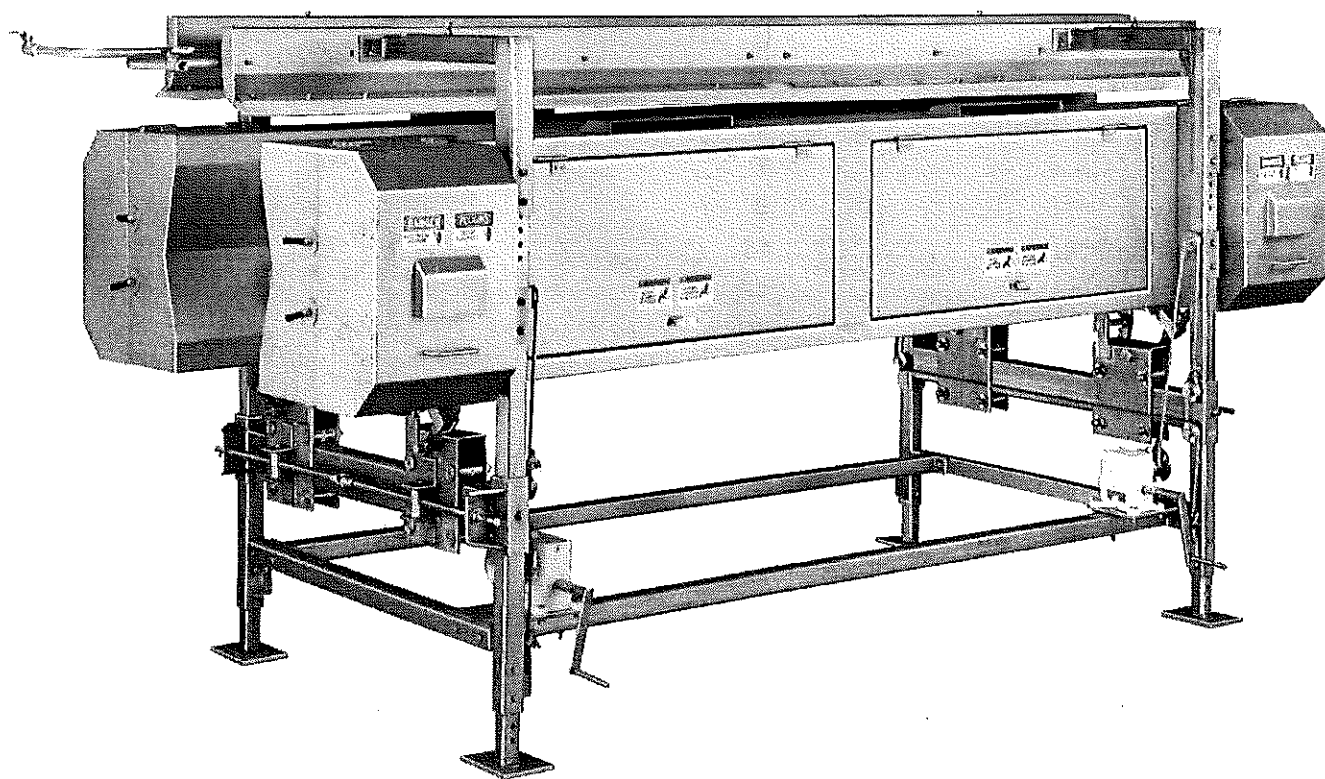


Meyn

Picking Machines



Model HM01

Meyn America LLC, 4052 Thurmon Tanner Parkway, Flowery Branch, Georgia 30542, USA

Table of Contents

Introduction.....	3
General Description.....	3
Safety Precautions.....	4
Technical Specifications.....	8
Warranty.....	7
Installation.....	9
Initial Adjustments.....	11
Operating Adjustments.....	12
Maintenance.....	13
Replacing Fingers.....	14
Replacing Bearing Units.....	14
Replacing Idler Pulley.....	14
Trouble Shooting.....	15
Replacement Parts.....	16
Maintenance Log.....	28
Lubrications Log.....	29

Introduction

General Description

The HM01 picker is fully automatic and works in conjunction with the overhead conveyor system.

The shackle in which the bird hangs, is positioned between the shackle guide and is fed into the machine with the aid of the overhead conveyor.

Proper de-feathering of the bird is achieved by the rotation of the picking discs. The picking discs are cleaned with a constant supply of water from the spray pipes.

The stainless steel cabinets that house the bearing units with discs are mounted to a stainless steel stand. The stand is easy to adjust and locks into place for safe performance.

Safety Precautions

As with any operating machinery, safety precautions should be observed.

- **Read, Understand and Practice** the safety and operating instructions in this manual.
- **Hand Safety:** Keep hands outside the machine. Never reach into the machine while it is in operation.
- **Eye Safety:** Wear an approved safety shield, goggles or glasses to protect the eyes when cleaning or performing maintenance on this machine.
- **Do not** stand on, climb on or lean over the machine while in operation.
- **Lock out-Tag out!** When performing maintenance on any machine, make sure the switch is off, the electrical controls in the central control area have been turned off and locked out. Make sure there is a **Do Not Operate** tag in place on the machine.
- **Personal Safety:** Before performing any maintenance on the machine, remove loose clothing and accessories such as ties, rings, watches and jewelry. Roll sleeves above the elbows. Tie back hair. Protective footwear is suggested. Make sure that all persons working with or near the machine understand the potential dangers associated with working on machinery.
- **Always** give your work your undivided attention. Do not become distracted by talking, looking, or playing around while operating, cleaning or performing maintenance on the machine.
- Keep all access covers in place when power is applied. **Never** attempt operational adjustments, cleaning or removal of blockages while machine is in operation.
- **Electrical Connections:** Have all wiring done by a licensed electrician in accordance with national and local electrical codes. Provide one or more emergency cut-off switches, on or near the machine, in places that are easily accessible in case of accident.
- **Do not** use the machine for any purpose other than its intended use. If used for other purposes, Meyn disclaims any warranty, which might otherwise have applied in accordance with Meyn's standard limited warranty, and holds itself blameless for any injury or damage which might result from that use.
- **Replacement Parts:** Use only Meyn or factory authorized replacement parts and accessories.
- **Refer** all operation, maintenance or service to qualified personnel. Production or clean up personnel should never perform maintenance.
- Know and understand the **safety labels** on the equipment. Periodically you may need to replace them if they are damaged or lost.
- Adjust the machine to the **fully opened position** when performing maintenance.
- **Secure all cabinet doors** properly during maintenance. **Do not** prop open motor doors during operation.
- **Cover all exposed electrical connections** to prevent damage to equipment or personal injury.
- **Always** make sure all control buttons and switches work properly before operating the machine.
- **Do not** rely solely on the safety guards, emergency stop buttons or other safety devices provided for your machine.

Safety Precautions

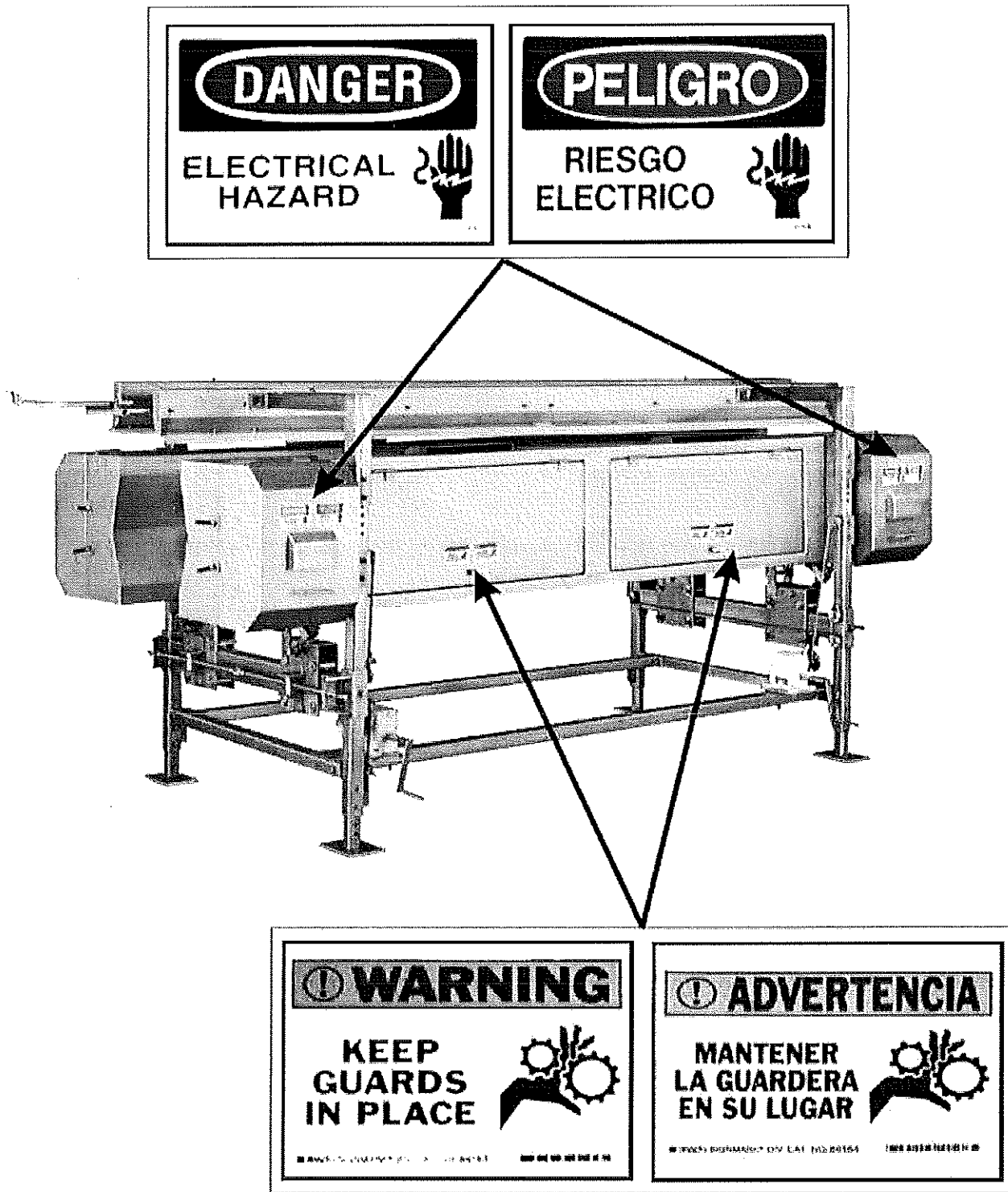
- **Do not** allow fatty or other slippery deposits to build up on the floor around the machine.
- **Do not** operate, clean or perform maintenance on the machine if you are not feeling well, taking over-the counter, prescription medicine or any other substance that may impair judgment.
- **Always** report any problem with your machine to your supervisor immediately.

●

● **Disclaimers:**

- There are no manufacturers warranties on this stated machine except as expressed in this owners manual.
- If this machine is used for any purpose other than it was intended by Meyn, the manufacturer disclaims any warranty, which might otherwise have applied in accordance with Meyn's standard limited warranty, and holds itself blameless for any injury which might result from such misuse.
- Unauthorized repairs—Meyn holds itself blameless for any injury or damage arising or resulting from repairs, alterations, modifications or adjustments not specifically authorized by this owner's manual or our written authorization including installation by persons not employed by Meyn.
- Replacement parts—use only Meyn parts or Factory authorized replacement parts and accessories. Otherwise, Meyn disclaims any warranty and holds itself blameless for any damage.

Placement of Safety Alerts



Warranty

Technical Specifications

Drive: 8 electric motors: 3hp, 3phase, 60 hertz, 1740 rpm, 208-230/460 volts, 7/8" shaft.

Water: Two 3/4" NPT supply lines, 40 PSI minimum.

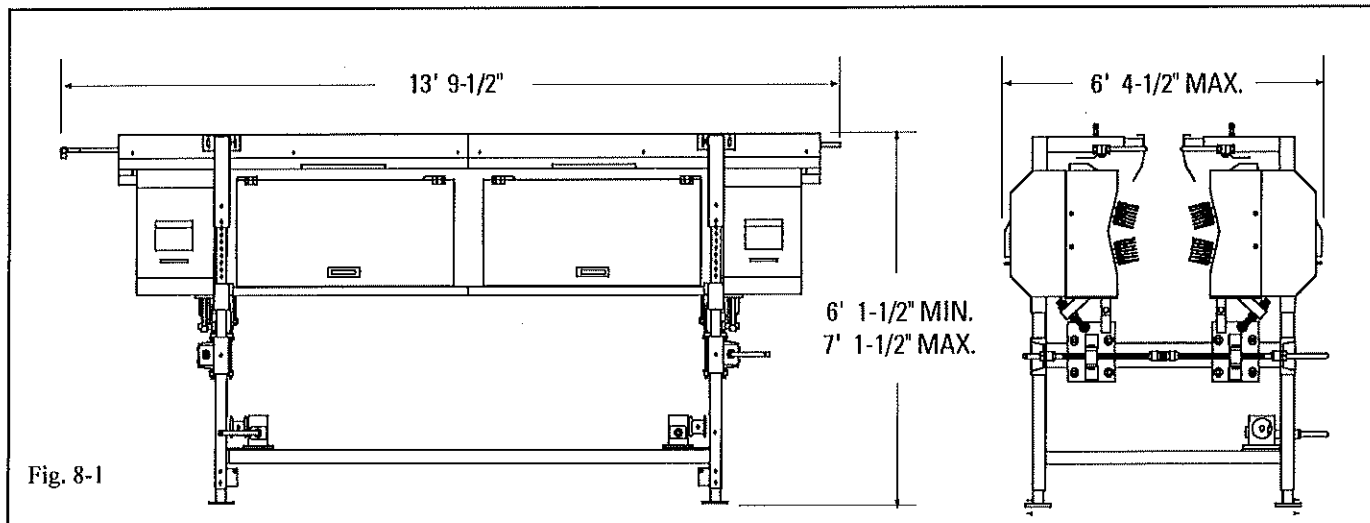
Weight: 3800 lbs.

Height: 6'-1 1/2" Minimum, 7'-1/2" Maximum

Length: 13'-9 1/2"

Width: 6'-4 1/2" Maximum

Dimensions



Installation

Requirements:

Supply Lines:

Water: 3/4" NPT supply lines, 40 PSI minimum, 20 PPM chlorine

Electrical: 60 hertz, 208-230/460 volts.

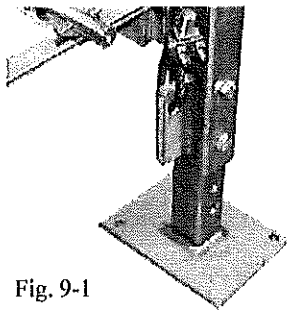


Fig. 9-1

Installation Procedure:

Note: Water and electrical supply lines are required for installation. Electrical installation and repairs are to be performed only by a licensed electrician in accordance with manufacturer's specifications, national and local electrical codes. It is also recommended that spare parts be on hand at the time of installation.

Positioning: If the picker is in the first position, place it as close as possible behind the scalders. If it is used in any position other than first, place it as close as possible to the preceding picker and parallel to the overhead conveyor. The conveyor should run horizontally over the picker.

Leveling: Initial leveling of the machine can be done using the two bolts located near the bottom of the legs on the frame. Adjust the height of each end by removing the two bolts and replacing in pre-drilled holes. Fine adjustment is done with the cabling system. Level machine and lock in place. See Fig. 9-1.

Height Adjustment: Initial set up is approximately 6-5/16" from the top of the cabinet to the bottom of the shackle. See Fig. 9-2.

Shackle Guide: Adjust the spray pipes to approximately 7/8" apart. The shackle should hang straight down ward between the guides. Adjust in as necessary to prevent unnecessary motion. See Fig. 9-3

Water Connection: Connect the spray pipes to the water supply lines. Water temperature should not exceed 115° Fahrenheit or 46° Celsius .

Electrical Connection: Before connecting the motors, check whether the available voltage and frequency are the same as those stated on the motor specification plates. The motor plate should state: 3 phase, 60 hertz, 208-230/460 volts.

Disc Rotation: The direction of rotation of the first finger disc can be seen in Fig. 10-1. The direction of rotation of the following finger disc is always the opposite direction of the preceding disc.

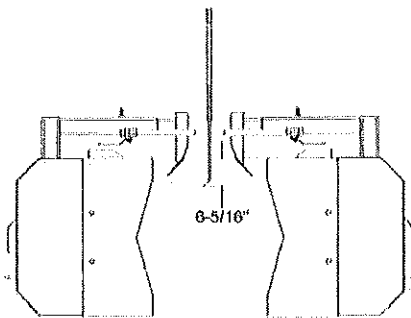


Fig. 9-2

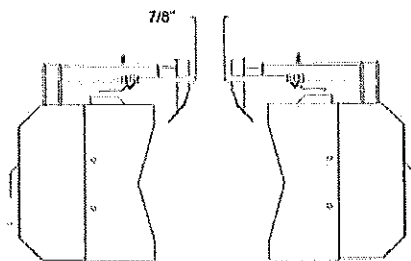
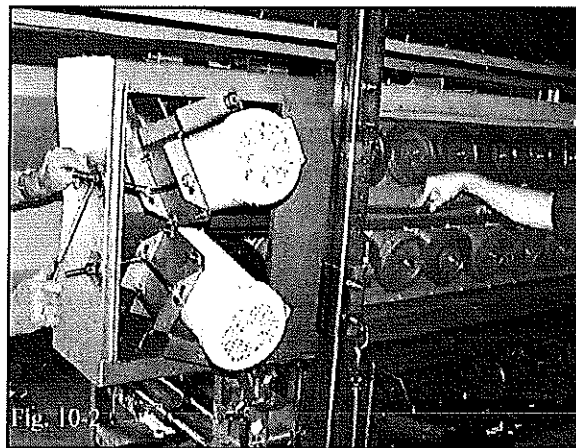
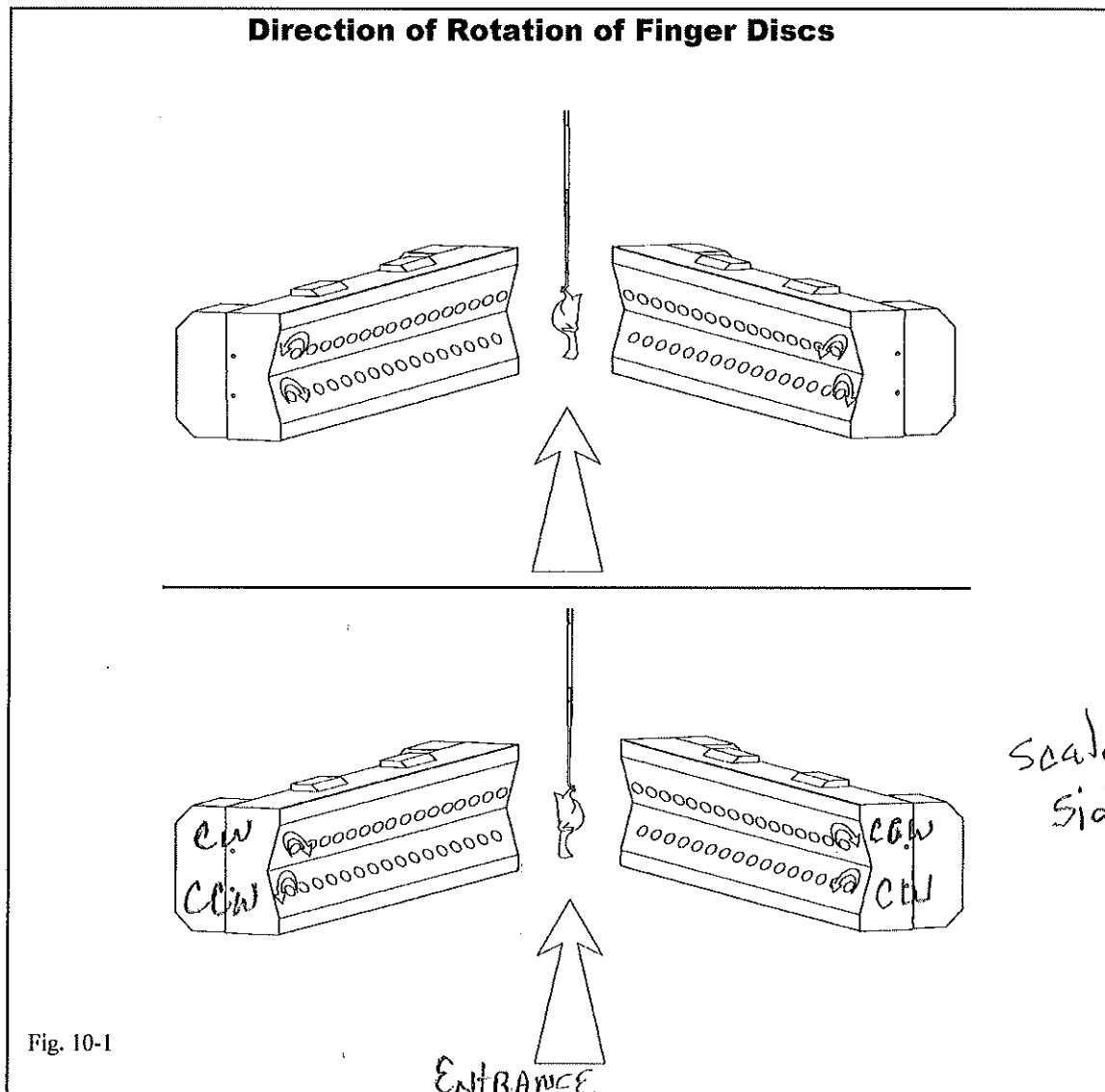


Fig. 9-3

Installation



Initial Adjustments

1. Before starting the picker:

- A. Check to make sure shackles are suitable for the picking operation of the machine. If not, replace.
- B. Check all water and electrical connections. Make sure they have been properly made and that they meet the minimum requirements.
- C. Check the belts and make sure they are seated properly on the pulleys.
- D. Check to make sure all lock nuts, bolts, and set screws are tight.
- E. Check to make sure machine is clear of all workers, tools or other objects.

1. When installing or adjusting the belts:

- A. **Do Not** over-tighten. This contributes to early belt and bearing wear. Tighten belts just enough to prevent belts from slipping on the pulleys during operation. For the first couple of days after installing new belts, minor tensioning may be required due to stretching.
- B. Initial belt tension is achieved when 2 -1/2 to 3 lbs. force deflects the belts 3/4". Deflection should be measured in the middle of the section of belt between the idler and the motor. Adjust the belt tension at the motor mount.. See Fig. 10-2.
- C. The belt tension is sufficient when the finger disc revolutions are approximately 960 RPM during production. This should be measured with a tachometer. The belt has to make several revolutions during the tensioning process in order to distribute the tension in the belt equally.
- D. The distance between the face of the idler pulley and picker cabinet is approximately 4-3/8".

Operating Adjustments

Slight adjustments are periodically required when setting up a new machine. Also adjustment may be required when bird sizes change.

Always make a final inspection of the machine before starting to ensure that:

1. All adjustments have been made.
2. All lock nuts, bolts and set screws are tight.
3. All supply lines are connected and turned on and meet the minimum requirements.
4. The machine is clear of all tools and other objects.

Operation

1. Engage the motors
2. Open the water supply
3. Adjust the picker, by cranking to the correct width for the product being picked. Never adjust the picker too tight; this will cause product damage and possibly discoloration or bruising. Following are some suggestions for initial set up:
 - a)Broilers: Cabinet width at infeed: 8.5", cabinet width at outfeed: 7.5"
 - b)Layers: Cabinet width at infeed: 9", cabinet width at outfeed: 8.75"
4. The top row of the rotating fingers on the side banks should be positioned so they almost touch the bottom of the shackle when the machine is running. The picker banks should be kept level from the infeed end to the outfeed end and parallel with the overhead conveyor.
5. Tilt adjustment on the side banks can be adjusted to put emphasis on the hocks or the wings, depending on the need.

Maintenance

Lubrication / Maintenance of Bearing Housing Assembly:

Daily :

1. Check the picking fingers for wear and replace as necessary.
2. Clean the outside of the machine everyday with warm water. During the cleaning cycle, the finger discs should be rotating and the covers closed.
3. Check belts for wear and proper tension.
4. Inspect cabinets and pulleys for signs of grease.

Weekly:

1. Check the nuts and bolts.
2. Check the working order of adjustments.
3. Check bearings for wear.
4. Check idler pulleys bearings.

Monthly:

*Seller recommends lubricating each bearing housing assembly **ONCE PER MONTH** with **ONE SHOT** of grease from a standard grease gun. Seller also requires recording monthly maintenance using the Lubrication Log on page 29.*

Maintenance

It is recommended that you remove doors/covers before performing maintenance for the safety of personnel.

Replacing Fingers

Broken or worn picking fingers should be replaced by new ones to avoid damage to the birds while being processed and to ensure proper picking is achieved. First, Lock Out/Tag Out the machine to make sure it is not accidentally started up during replacement of the fingers. To change the picking fingers, adjust the width of the cabinets to the fully open position. Cut the damaged finger from the disc, wet the replacement and pull it into place on the disc. Store the picking fingers in a dark, dry place as recommended by the manufacturer.

Removing and Replacing Bearing Units

1. Cut power
2. Loosen motor
3. Remove finger disc
4. Remove bolts
5. Remove bearing unit from picker cabinet.
6. Replace bearing unit.
7. Insert bolts and tighten.
8. Replace finger disc.
9. Tighten motor.

Removing and Replacing Idler Pulley

1. Cut power
2. Loosen motor
3. Remove two bolts
4. Remove idler unit from picker cabinet.
5. Replace idler unit .
6. Insert bolts and tighten.
7. Tighten motor.

Troubleshooting

Regular maintenance and inspection is key to avoiding costly down time for repairs. Following is a guide for trouble shooting:

Problem: Inefficient Picking

Possible Causes:

- Width adjustment is too wide – adjust width
- Rotation of the finger discs is wrong – check rotation diagram in Initial Adjustment section
- Inefficient scalding – check scalding specifications
- Picking fingers are worn or broken – replace fingers as necessary
- Finger discs are slipping – increase belt tension, check for oil/lubrication on pulleys.

Problem: Product damage

Possible Causes:

- Width adjustment is too narrow – adjust width
- Shackles may be damaged – inspect shackles and bird placement in shackle stirrups
- Rotation of the finger discs is wrong – check rotation diagram in Initial Adjustment section

Problem: Excessive heating of belts and pulleys.

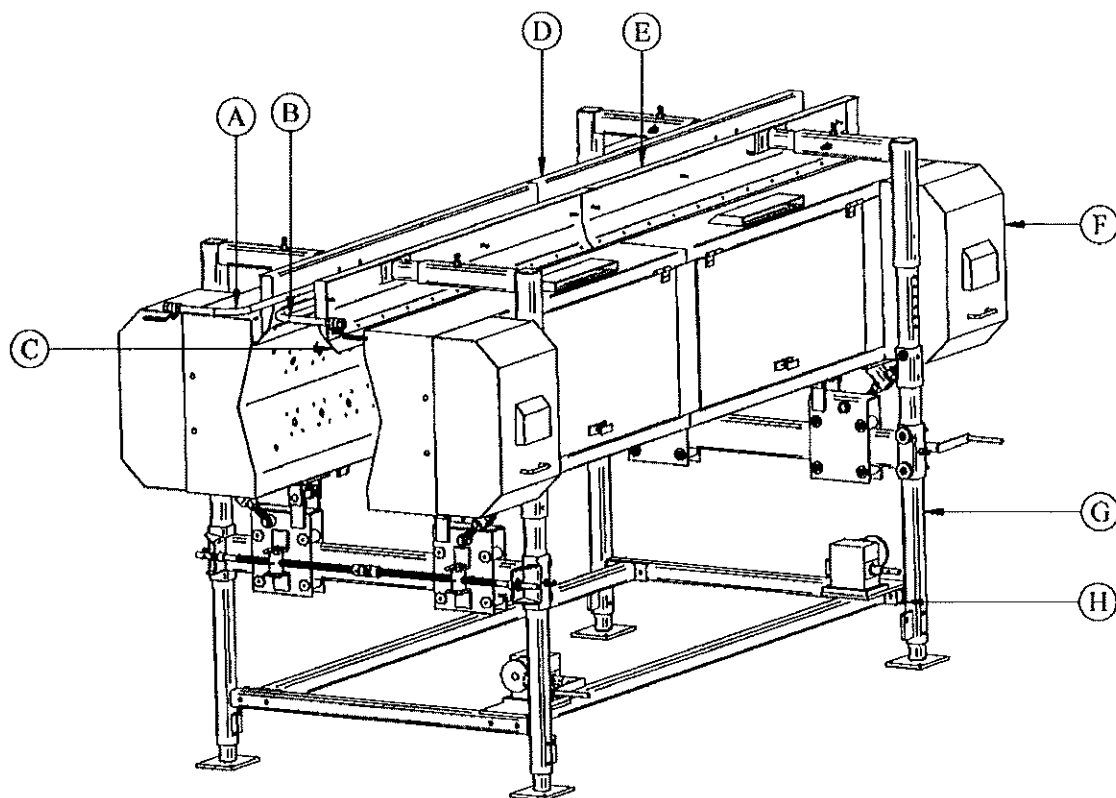
Possible Causes:

- Belt too tight—tension belts properly.
- Belt too loose— tension belts properly

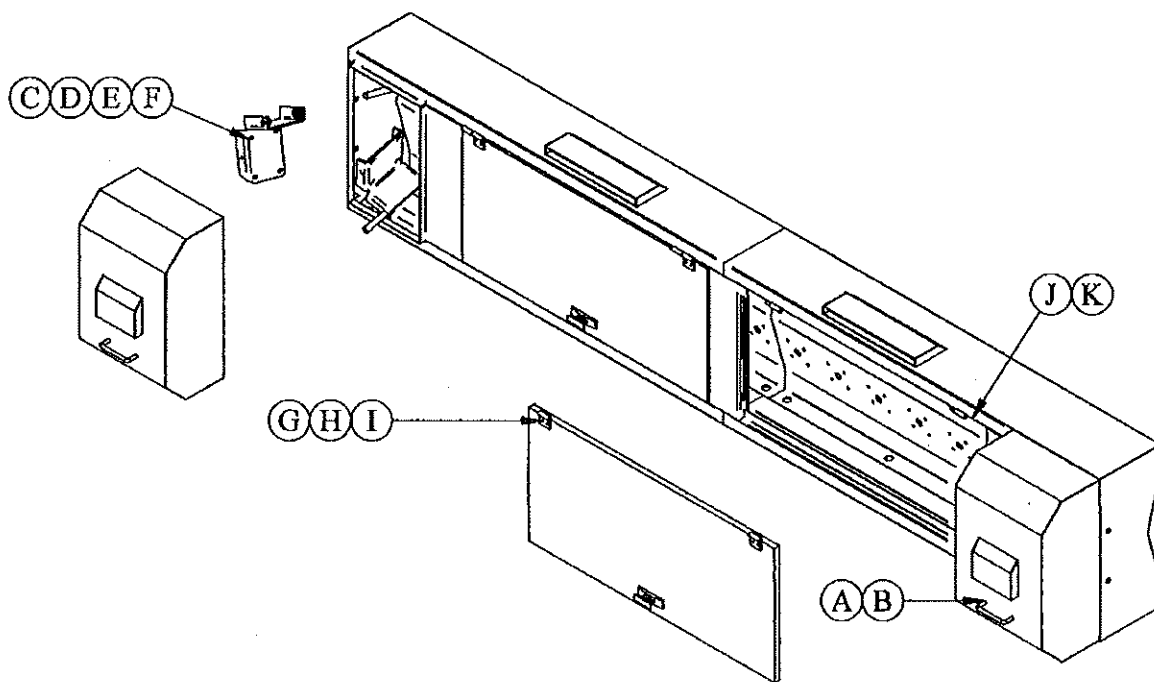
Recommended Spare Parts List

Qty	Part #	Description
2	MYPK-810	Idler Assembly
4	401213	Replacement Bearings for Idler
4	401452	"V" Belt
6	MYPK-820	Housing Assembly with Pulley
1	GSPK-550	3 HP Motor
2	MYPK-707	12 Hole SS Finger Disc

203119
11-2100

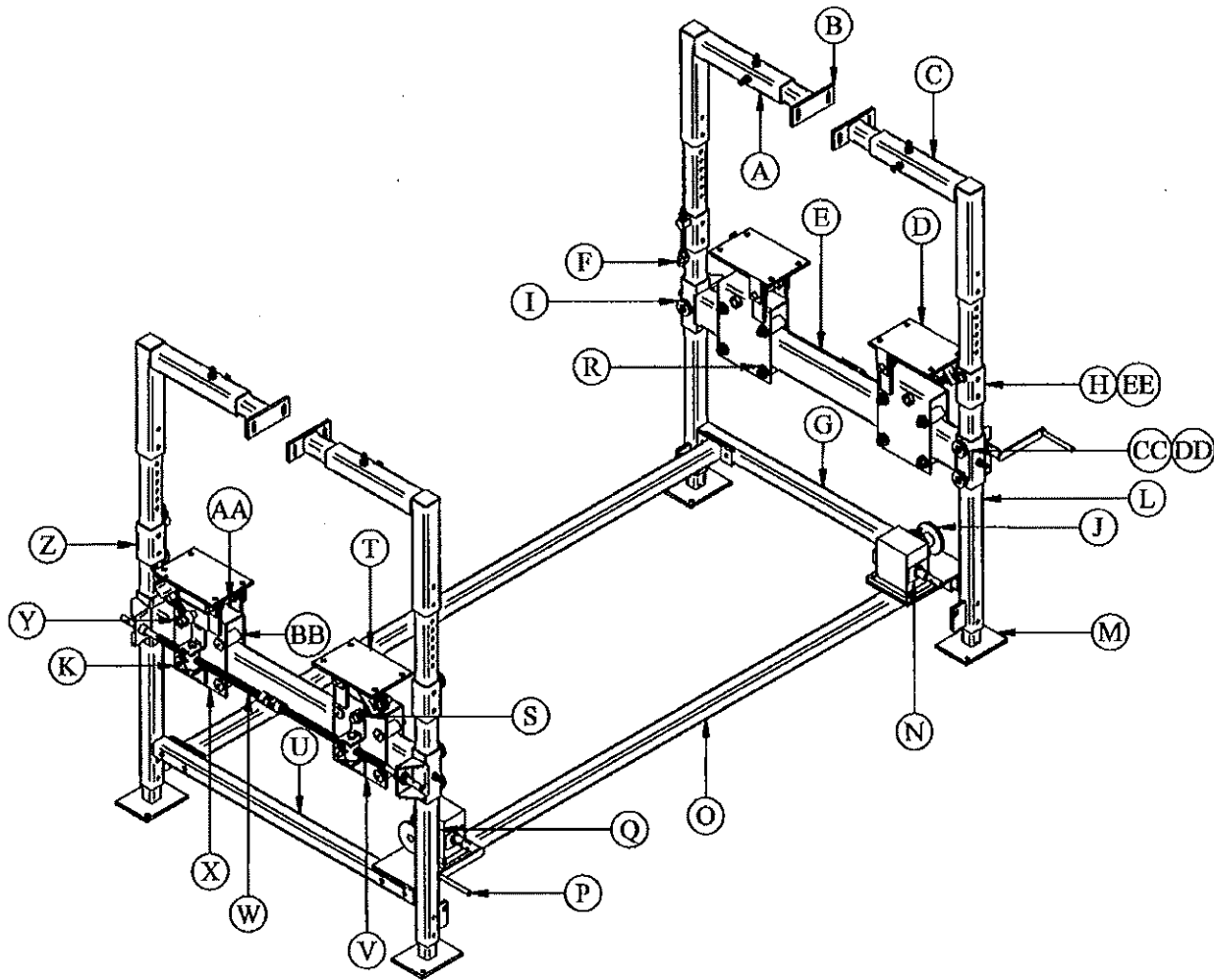


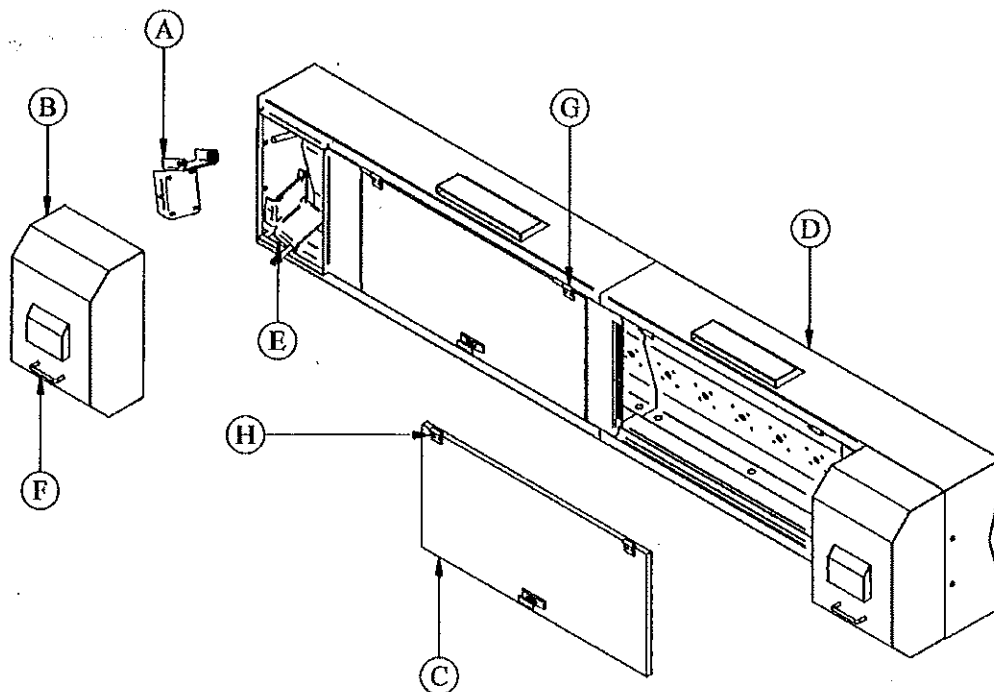
Cabinet Assembly			
Location	Part Number	Description	Qty Req
A	GSPK001-4LH	LH Spray Bar Assembly	1
B	GSPK001-4	RH Spray Bar Assembly	1
C	GSPK001-3	Splash Guard	2
D	GSPK001-5LH	LH Spray Bar Shield Assembly	1
E	GSPK001-5	RH Spray Bar Shield Assembly	1
F	GSPK005	Cabinet Shell Only for 32 Unit VB164	2
G	HLPK004	Narrow Picker Stand Assembly	1
H	HLPK004A	8" Shim	4



Cabinet Hardware			
Location	Part Number	Description	Qty Req
A	4050-001	1/4-20 x 5/8" SS Hex Head Cap Screw	8
B	4050-005	1/4 SS Lock Washer	8
C	405052	Flat Washer	8
D	405047	Lock Washer	4
E	405064	Nut	4
F	405051	Bolt	4
G	4050-002	1/4-20 x 3/4" SS Hex Head Cap Screw	16
H	4050-004	1/4 SS Flat Washer	16
I	4050-003	1/4-20 SS Lock Nut	16
J	405009	M6x16 SS Hex Head Cap Screw	8
K	405021	M6 SS Nylon Insert Lock Nut	8

Frame Components

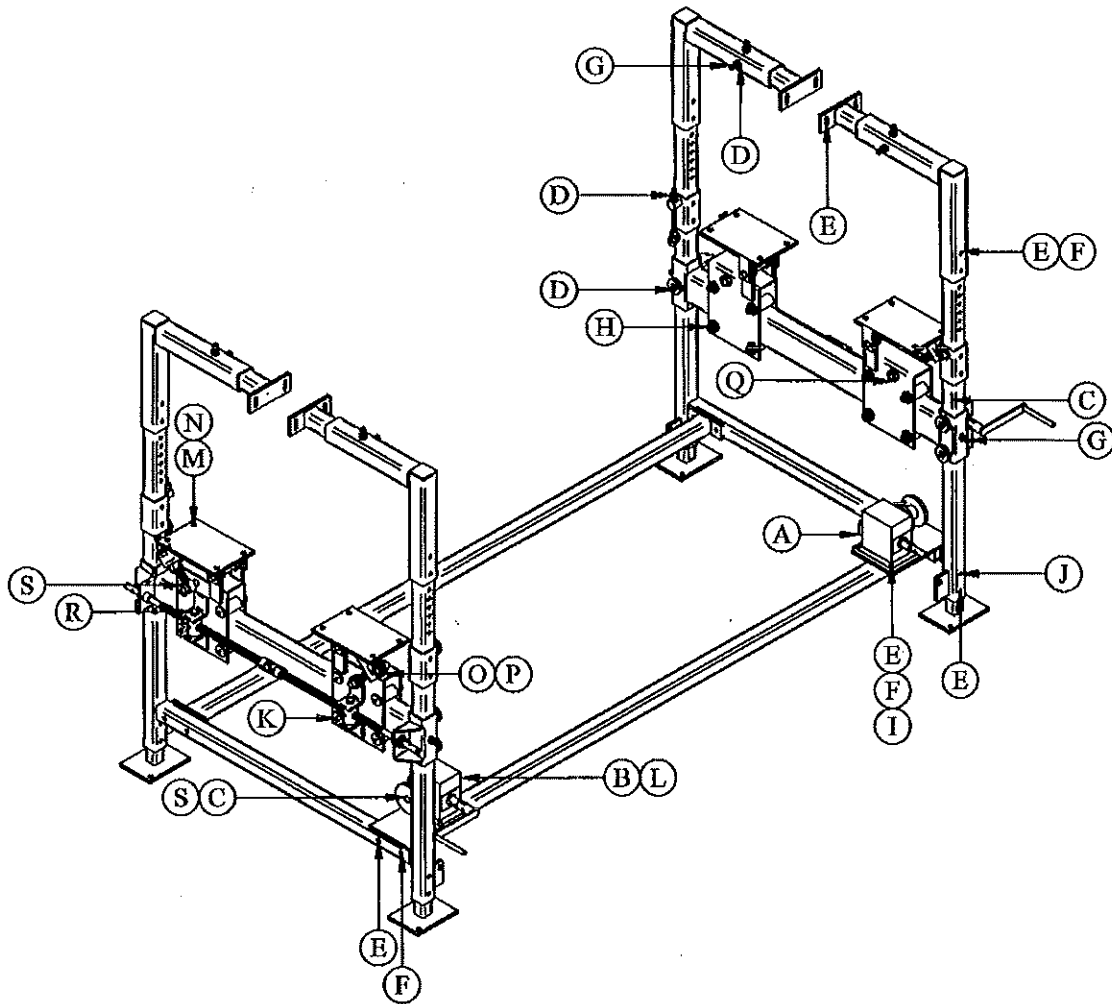




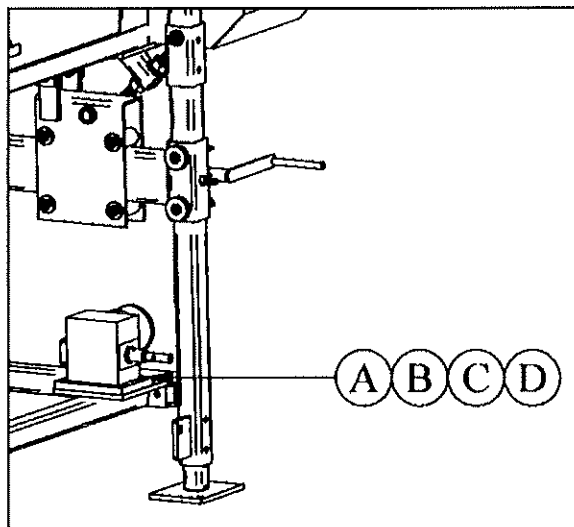
Cabinet Components			
Location	Part Number	Description	Qty Req
A	GSPK005-1	Motor Mount Assembly	2
B	GSPK005-2	Motor Door	2
C	GSPK001A	Cabinet Door	2
D	GSPK005K	Fabricated Cabinet Assembly	1
E	HLPK001F	Sandwich Strip, cabinet mount	4
F	GSPK005-2C	Motor Door Handle	4
G	GSPK001-1D	RH Hinge Receiver	4
	GSPK001-1E	LH Hinge Receiver	4
H	GSPK001-1A	Hinge Shaft	4

Frame Components			
Location	Part Number	Description	Qty Req
A	HLPK0004-6	RH Adjustable Support Arm for Spray Bar Shield	2
B	HLPK0004-6J	Adjustable Spray Bar Shield Bracket	4
C	HLPK0004-6L	LH Adjustable Support Arm for Spray Bar Shield	2
D	HLPK0004-5L	LH Cabinet Tilt Plate	2
E	HLPK0004-4	Horizontal Cable Slide Assembly	2
F	HLPK0001-13	Eye/Nut Assembly	2
G	HLPK0004-1SR	Lower Cross Support and Winch Mount RH	1
H	HLPK0004-7R	RH Cable Tie-off Assembly	2
I	HLPK0001C	SS Cable Pulley	8
J	HLPK0001-10	Gear Box Pulley	2
K	HLPK0001-2G	Bolt for Roller	16
L	HLPK0004-3R	SS Leg Assembly, RH	2
M	HLPK0001-9	Adjustable Leg Assembly	4
N	HLPK0001-11	Gear Box Handle Adapter	2
O	HLPK0004-15	Horizontal Stabilizer Tube	2
P	HLPK0001-12	Adjustment Handle	2
Q	HLPK0004-3L	SS Leg Assembly LH	2
R	HLPK0001-2H	Washer for Roller (Cabinet Side)	16
S	HLPK0004-14R	RH Tilt Adjustment Screw	4
T	HLPK0004-5R	RH Cabinet Tilt Plate	2
U	HLPK0004-1SL	Lower Cross Support and Winch Mount LH	1
V	HLPK0004-2RH	RH Horizontal Cabinet Slide Assembly	2
W	HLPK0004-8	Saddle Adjustment Rod Assembly	2
X	HLPK0004-2LH	LH Horizontal Cabinet Slide Assembly	2
Y	HLPK0004B	Tilt Adjustment Pivot Pin	4
Z	HLPK0004-7L	LH Cable Tie-off Assembly	2
AA	HLPK0001J	Pivot Pin for Cable Adjustment Mounting Plate	4
BB	HLPK0001-2F	Roller for Horizontal Cabinet Slide Assembly	16
CC	HLPK0001B1	Vertical Cable with Stop	2
DD	HLPK0001B2	Horizontal Cable with Stop	2
EE	HLPK0001K	"L" Shim	8

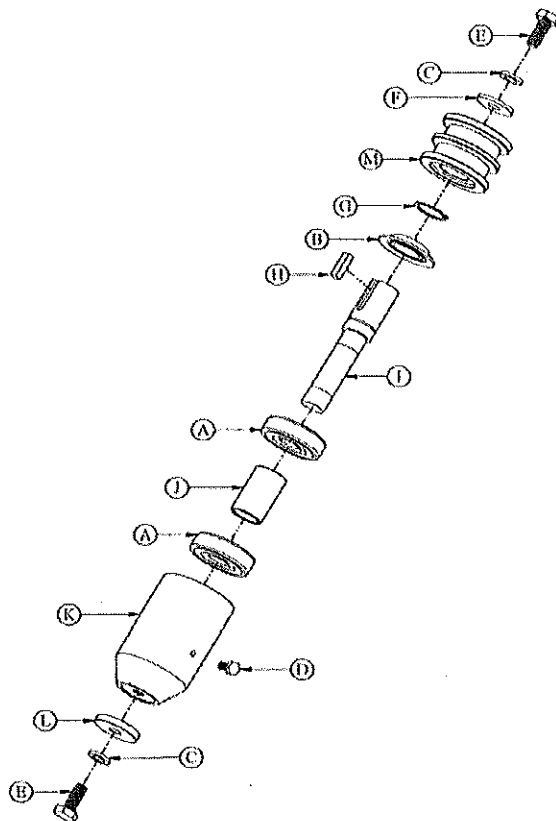
Frame Hardware



Frame Hardware			
Location	Part Number	Description	Qty Req
A	4054-092	Gear Reducer, RH	1
B	4054-091	Gear Reducer, LH	1
C	4054-078	3/8 SS Cable Clamp	8
D	4050-070	1/2-13 SS Hex Nut	20
E	4050-071	1/2-13 SS Nylon Lock Nut	50
F	4050-136	1/2-13 x 4" SS Hex Head Bolt	26
G	4050-121	1/2-13 x 1 1/4" SS Sq. Hd. Cup Pt. Set Screw	12
H	4050-112	5/8-11 SS Nylon Insert Hex Lock Nut	16
I	4050-126	1" SS Split Lock Washer	4
J	4050-093	1/2-13 x 3 1/2" SS Hex Head Cap Screw	8
K	4054-081	3/16 SS Press In Grease Fitting	20
L	405432	5/16-18 x 1/4 SS Cup Pt. Set Screw	16
M	4050-152	1/2-13 x 1 3/4 SS Hex Head Cap Screw	16
N	405060	1/2" SS Lock Washer	16
O	4050-127	1" SS Flat Washer	8
P	4050-124	1"-8 SS Hex Nut	8
Q	4054-072	1" SS Set Collar	12
R	4050-123M	Bolt for Threaded Blocks (5/8-11)	8
S	4054-087	1/4 x 1 1/2" SS Key	2

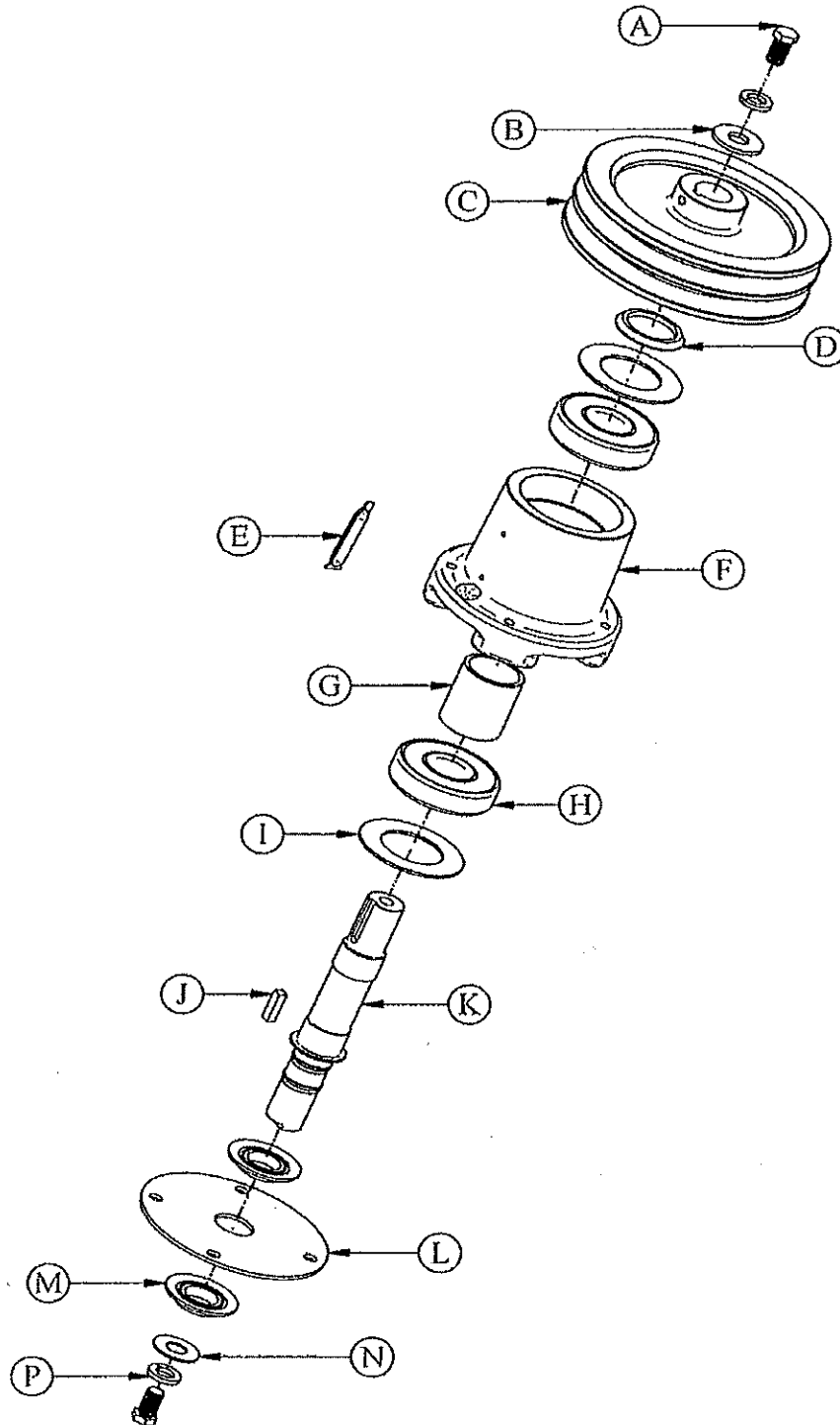


Location	Part Number	Description	Qty Req
A	4050-040	7/16-14 SS Hex Nut	8
B	4050-041	7/16 SS Lock Washer	8
C	4050-051	7/16-14 x 1 3/4" SS Thin Hex Hd. Cap Screw	8
D	405074	7/16 SS Flat Washer	8



MYPK-810 Idler Assembly				
Location	Part Number	Description	Qty Req	
A	401213	Bearing	2	
B	401305	Water Slinger	1	
C	4050-041	Lock Washer	2	
D	405009	Hex Head Cap Screw	1	
E	405011	Hex Head Cap Screw	2	
F	405023	Flat Washer	1	
G	405425	Retaining Ring	1	
H	405479	Key	1	
I	GSPK-710A	Shaft	1	
J	GSPK-710B	Bearing Spacer	1	
K	GSPK-710C	Bearing Housing	1	
L	GSPK-710D	Idler Washer	1	
M	GSPK-710E	Idler Pulley	1	

Picker Bearing Assembly



MYPK-820 Picker Bearing Assembly Complete			
Location	Part Number	Description	Qty Req
A	4050-107	Hex Head Cap Screw	2
B	4050-194	Washer	1
C	MYPK-808M	Modified Pulley	1
D	401334M	Modified Water Slinger	1
E	LUB-SSP1212-1ML	Lube Packet	1
F	MYPK-803	Housing	1
G	GSPK-705	Bearing Spacer	1
H	401290	Bearing	2
I	GSPK-701	Thrust Washer	2
J	405479	Key	1
K	MYPK-802	Shaft	1
L	MYPK-809	Isolator Plate	1
M	401305	Water Slinger	2
N	405062	Washer	1
P	405060	Lock Washer	2

Lubrication Log

Lubrication Schedule - One (1) Shot Per Month

Maintenance Person	Date Maintenance Performed			Notes
	Month	Day	Year	
	Jan			
	Feb			
	Mar			
	Apr			
	May			
	Jun			
	Jul			
	Aug			
	Sept			
	Oct			
	Nov			
	Dec			
	Jan			
	Feb			
	Mar			
	Apr			
	May			
	Jun			
	Jul			
	Aug			
	Sept			
	Oct			
	Nov			
	Dec			

Attention Maintenance Personnel:

Please fill in the day and year each time lubrication is applied.

Lubrication Log

Lubrication Schedule - One (1) Shot Per Month

Maintenance Person	Date Maintenance Performed			Notes
	Month	Day	Year	
	Jan			
	Feb			
	Mar			
	Apr			
	May			
	Jun			
	Jul			
	Aug			
	Sept			
	Oct			
	Nov			
	Dec			
	Jan			
	Feb			
	Mar			
	Apr			
	May			
	Jun			
	Jul			
	Aug			
	Sept			
	Oct			
	Nov			
	Dec			

Attention Maintenance Personnel:

Please fill in the day and year each time lubrication is applied.

